

Introduction to High-Performance and Parallel...Computing

Today's goals

Module 1

High-Performance Computing (HPC) for Non-Computer Scientists

Module 2

Nuts and Bolts of HPC

Module 3

Basic Parallelism

Module 4

Evaluating Parallel Program Performance

How to Parallelize Code

How to Parallelize Code
Video • 6 min

Speedup and Parallel Efficiency

Speedup and Parallel Efficiency
Video • 4 min

Scalability

Scalability
Video • 4 min

Limits to Parallel Performance

Limits to Scaling
Video • 3 min

Module Assignments

Week 4 Quiz

Graded Assignment • Grade: 100%

Strong Scaling Study

Programming Assignment • 1h

Weak Scaling Study

Programming Assignment • 1h

Course Summary

Summary of This Course

Reading • 10 min

Programming assignment

Weak Scaling Study

This will start your JupyterLab environment. Work this week in the directory "projects/week4-scaling"

InstructionsMy submissionsDiscussions

In this exercise, you will perform a weak scaling study using an existing program.

In your JupyterLab environment, you have four files in your `/projects/week4-weak_scaling/` directory: `scale.F90`, `Makerfile`, `weak_input.dat`, and `weak_scaling.sh`.

To begin, compile the Fortran90 program by loading either the `mpich` or `openmpi` module in the terminal and typing `make`. This will generate an executable named `scale.fexe`.

`scale.fexe` performs a smoothing operation on a 2-D grid, multiple times and in parallel. The calling syntax is:

```
mpiexec -np N ./scale.out -nx X -ny Y -nt T
```

where the following command-line parameters must be specified

- N: number of MPI ranks (number of processors)
- X: number of gridpoints in x-direction (default = 1024)
- Y: number of gridpoints in y-direction (default = 4096)
- T: number of iterations to run for (default = 100)

The Y gridpoints are distributed across the N processors. This means that Y should be at least N and that (ideally) N should divide evenly into Y. Each process contains all X gridpoints in the X-direction; those are not distributed.

Note that NY is the problem size and T is the computation size. Rewatch Lecture 3 to review how the problem size and computation size scale with increasing number of processors in a weak scaling study.

`scale.fexe` is called in the `weak_scaling.sh` bash script using the `-nx -ny` and `-nt` parameters in the `weak_input.dat` file.

The first condition has been written in the input file for you; this condition is for 1 processor (N) and X=1000, Y=10,000, and T=900. Add two more conditions for 2 and 4 processors to the input file for a total of three conditions.

Once you have completed your input file, perform the weak scaling study by running the bash script. When the program completes, it will print a single line of text indicating the problem parameters and time, separated by whitespaces, formatted as N time X Y T, for each condition in the input file.

Once you are satisfied with your study, submit the `weak_input.dat` file for autograding. The autograder will examine the three `-nx X -ny Y -nt T` lines in your submission.



What to expect

- Due Sep 21, 11:59 PM IST
- Unlimited attempts

Report an issue

Go to next item →